

RITIK AGARWAL

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WORK EXPERIENCE

Independent Blockchain & Smart Contract Security Researcher

Jan 2025 - Present

- Completed a **private security audit** of a **Web3 infrastructure protocol** under NDA; identified and reported multiple valid vulnerabilities.
- Delivered **18 validated security findings** across a broad range of DeFi protocol types, including automated market makers, lending and borrowing markets, ERC-4626 vault architectures, and liquidity management systems.

PERSONAL PROJECTS

Shim - Co-Founder ([Live](#))

May 2026 - Present

- Built and shipped **Shim**, a **Go-based security proxy** for npm, pnpm, and yarn that **recursively scans direct and transitive** (sub-package) **dependencies** for known vulnerabilities before install.
- Engineered a daemon scanner using **TCP IPC**, **local vulnerability rules**, **lifecycle-script analysis**, and **SQLite-backed caching**, adding minimal overhead compared to native npm installs.
- Shipped **cross-platform** release tooling (install/update/uninstall) for Linux, macOS, and Windows, reaching **25+ early users** since launch.

Multisig Smart Contract Wallet - Account Abstraction

Feb 2024 - Mar 2024

- Built a **multisig smart contract wallet** using ERC-4337 account abstraction, requiring multiple designated signers to jointly authorize and approve a transaction before it executes on-chain.
- Implemented wallet creation, **signer management**, and the **propose-approve-execute** transaction flow using Solidity smart contracts, OpenZeppelin libraries, and ethers.js.
- Built the supporting frontend and backend with **Next.js**, **Prisma**, and **Supabase**, with Alchemy providing blockchain RPC infrastructure.

Uniswap V4 Take-Profit Hooks ([GitHub](#))

Dec 2024 - Jan 2025

- Built custom Uniswap V4 hooks implementing **automated take-profit order execution**, integrating with the V4 core and periphery contract architecture.
- Developed and tested the hooks in Solidity using Foundry, validating execution logic against Uniswap V4's core and periphery contract interfaces.

TECHNICAL SKILLS

- Languages:** Solidity, Python, Go, SQL, JavaScript, C++
- Frameworks & Libraries:** Next.js, ethers.js, OpenZeppelin
- Databases & Tools:** Supabase, Prisma, SQLite, Git, GitHub
- Security & Audit Tooling:** Foundry (PoC development), Hardhat, Remix

EDUCATION

Jain University, Bangalore

Aug 2024 - Sept 2026

Master in Computer Application

Relevant Coursework: Network Security & Cryptography, Database Management Systems, Cloud Computing Foundations, Python Programming

GLS University, Ahmedabad

July 2021 - May 2024

Bachelor's in Computer Application - 84%

Lions English School, Silvassa

April 2019 - May 2021

High School, Commerce with Computer Science - 85%

LEADERSHIP & ACHIEVEMENTS

Kuru Smart Contract Security Review

Oct 2025

- Ranked 3rd of 723 participants**, with **4 validated findings**, in a smart contract security review.

Cyber Shadez Hackathon - Backend Developer

Feb 2024 - Mar 2024

- Led team to **2nd place**; built an **Account Abstraction DApp** using Supabase, Prisma, Alchemy, Next.js, Solidity, and OpenZeppelin.

Cyber Shadez Hackathon - Backend Developer

Feb 2023 - Mar 2023

- Led team to **1st place**; built and shipped an **NFT minting DApp** under competition conditions using OpenZeppelin, ethers.js, and Next.js.